Human-Rights-by-Design Engineering Strategies for AI-based Systems:

Protocol for Systematic Mapping Study

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Responsible	Description	Version
AA	First version of the protocol to be reviewed by the research group.	1.0
AA	Final protocol updated based on the suggestions of the steering group.	2.0
AA	Inclusion criteria updated based on the calibration exercise during the screening process.	3.0

1 Context, Scope and Contributions of the SMS

Context: In this study, our first goal is to map the human rights engineering strategies relevant to AI-based digital health systems and to do so in consultation with our business partners. Furthermore, this landscape mapping activity will help to provide more focus to our business partners, especially in identifying evidence-based human rights engineering strategies (i.e., having a proven record of validation or evaluation of empirical research). This is particularly important given the ever-increasing number of frameworks and assessment tools in the broader area of responsible AI. Although prior work has been done in this regard [1, 2], the reviews are not focused on digital health and, most importantly, lack adequate discussion on the maturity and proper evaluation of existing methods, i.e., existing strategies have not been adequately validated or evaluated in real-world settings. Moreover, while tensions between high-level ethical values are addressed in [1], conflicts between trustworthy AI protection goals and technical measures that are available for mitigating identified risks (e.g., see [3]) are not covered.

With that in mind, we adopted an Evidence-Based Software Engineering (EBSE) approach to conduct an extensive Systematic Mapping Study (SMS) [4] on the topic of Human-Rights-by-Design [5–8] for AI systems. To complement the SMS, we also performed an in-depth thematic analysis of the studies that have been evaluated in practice, discussing these solutions, their evaluation strategies, and impacts on industry-scale systems. To the best of our knowledge, there is no equivalent survey, systematic review, or mapping study of literature on this topic. Therefore, this SMS identifies, classifies, compares, and communicates existing research and its implications to relevant stakeholders (i.e., researchers, practitioners, policy-makers, healthcare providers, and broader society).

We also plan to refine the mapping of existing literature in a Human-Rights-by-Design taxonomy, i.e., human rights engineering approaches (techniques, methods, processes, frameworks, and tools), as the primary synthesis of results. The taxonomy can also consider prominent standards on AI risk management and algorithmic bias (e.g., NIST AI RMF, ISO 42001). The taxonomy will address how gender differences are articulated as a potential bias criterion. Hence, it can provide a holistic yet reflexive "engineering view" of human rights for AI in digital health. Thus, such a taxonomy helps us

identify research gaps, such as areas in the AI lifecycle that are under-researched or lack rigorously evaluated human rights engineering approaches, or gaps regarding the conflicts between trustworthy AI protection goals. Therefore, researchers and practitioners can benefit from a holistic mapping of the techniques, methods, processes, frameworks, and tools supporting Human-Rights-by-Design in digital health software engineering activities, further articulated by gender equality aspects.

Scope and Contributions: The primary objective of this SMS is to analyze the body of knowledge and provide an overview of the topic in terms of: 1) research and contribution types; 2) research trends and taxonomy; and, 3) challenges and solutions for Human-Rights-by-Design for AI systems; 4) critical appraisal of the existing empirical research. Bibliographical information and research trends also pinpoint predominant areas of research, under-researched areas, gaps, and future dimensions of research. Therefore, the main contributions of this SMS are:

- Classify and compare existing and emerging solutions for Human-Rights-by-Design for AI systems in the form of systematic maps, classification taxonomy, and illustrative trends.
- Evaluation focused analysis of the solutions - implemented in practice - to identify common themes and appraise the quality of these evaluation studies.

Research Impact: Empirical evidence, research, and development of Human-Rights-by-Design for AI systems are lacking, and research studies need to be carefully evaluated before academic solutions can be adopted or extended in an industrial context. The results of this SMS can be beneficial for:

- Researchers interested in quickly identifying the existing research can help formulate new hypotheses to be tested and propose innovative solutions for emerging challenges of Human-Rights-by-Design for AI systems.
- Practitioners who want to understand academic solutions in terms of architectural models, implementation strategies, and evaluation frameworks, etc., that could be adopted in an industrial context.

2 Research Method and Study Protocol

To plan, conduct, and document this SMS, we follow an evidence-based software engineering approach and specifically adhere to recommendations and guidelines to perform systematic reviews and mapping studies [4]. An illustrative view of the adopted research method is presented in Figure 1 that highlights three research phases, each comprising 2-3 tasks, and each phase has an outcome. For example, the initial phase, Planning the Mapping Study, comprises two tasks relating to (i) identification of the needs and (ii) specification of the research questions for the mapping study. The outcome of this phase is the scope and objectives of the SMS in terms of research questions that need to be investigated. SMS planning is the precondition for later phases of the methodology. By adopting well-known methodology from [4] as in Figure 1, we aim to strengthen the findings, support objective interpretation of results, minimize bias, and enable reproducible results. In the remainder of this section, we discuss phases of the research methodology.

2.1 Phase I – Planning the Mapping Study

2.1.1 Identify the Needs for SMS

To ensure that no prior survey, mapping study, or systematic review (i.e., secondary studies) has been conducted, we searched Scopus using a search string to locate any secondary studies on Human-Rights-by-Design for AI systems, as detailed in Listing 1. Scopus indexes journals and conference proceedings from major publishers, such as Elsevier, Springer, IEEE, ACM, and others. Based on Listing 1, we found four relevant secondary studies that addressed our 2.1.2, which are listed in Table 1. As follows, we briefly present the related work and discuss its limitations, motivating the need for the current SMS.

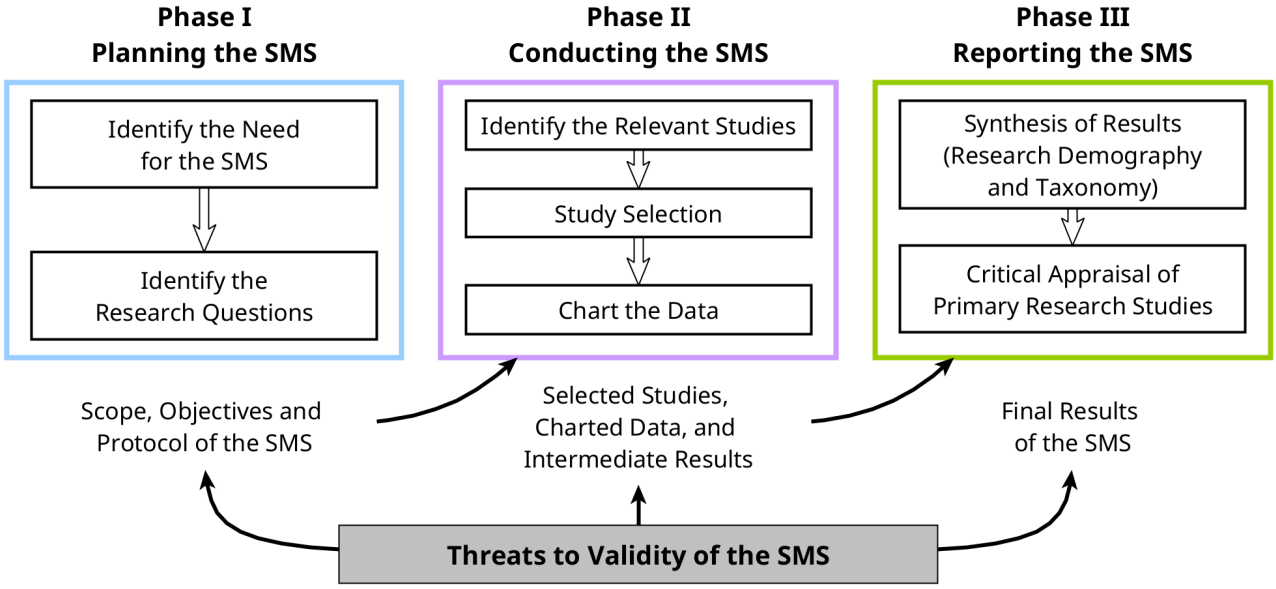

Figure 1: Methodology overview of the mapping process and evaluation-focused analyses.

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("mapping" OR "review" OR "survey") AND
(("human rights" OR "fundamental rights") AND
("artificial intelligence" OR "machine learning"))
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Listing 1: Search String to Identify Secondary Studies on Human-Rights-by-Design for AI Systems (August 2025).

As summarized in Table 1, none of the existing reviews precisely addresses our core research question on *human rights engineering strategies*. The work of [9] is mainly focused on identifying the intersection of human rights and AI as an *emerging research area*. Still, it did not intend to identify specific engineering methods.

The work of [1] addresses a broader area of AI Ethics, attempting to identify *tools*. However, the word "tools" is also used comprehensively, including regulations, guidelines, white papers, etc. These are not exactly engineering tools. Nonetheless, some tools they identified in the human rights area are relevant, such as the Human Rights, Social and Ethical Impact Assessment (HRESIA) proposed by [11]. We will therefore compare results between our SMS identified studies and the "tools" identified [1].

Similarly, the work of [2] focuses on risk assessment frameworks for the broad area of AI Ethics. Although they cover frameworks such as Fundamental Rights Impact Assessments (FRIAs), the scope of our study on engineering strategies aims to include more than just risk assessment methodologies. However, their findings have to be compared to ours, as we expect some overlap.

Lastly, the work of [10] offers insights from industry practices, emphasizing the human rights aspects of responsible AI. It is, however, important to differentiate this study from other systematic reviews of *scientific* literature. Even though the authors perform a systematic search and review of industry-authored resources in [10], these should not be confused with the systematic review of peer-reviewed scientific publications. Their searches are also not replicable, as companies' websites can change frequently. The listed best practices in the study are also mostly high-level, not pointing to specific engineering strategies (e.g., techniques, methods, processes, frameworks, and tools).

As a result of this preliminary search, we can motivate the need for our SMS, focusing on human rights engineering strategies. The related work is valuable for comparing findings and drawing conclusions with other overlapping areas. Our work, however, purposefully limits the scope to Human Rights instead of tackling broader (and perhaps vaguer) terms such as Responsible, Ethical, Trustworthy AI. Grounding AI development on human rights has concrete benefits in terms of their operationalization as advocated by [12]: (a) Unlike ethics (and codes of ethics), the International Human Rights Law (IHRL) is universal, offering a common vocabulary and set of principles that can be applied across

Table 1: Summary of existing surveys and systematic reviews.

Ref.	Year	SLR	Title	Limitations
[9]	2022	Yes	Artificial Intelligence and Human Rights: Are There Signs of an Emerging Discipline? A Systematic Review	**(i)** Focus on identifying AI & HR as an emerging discipline; not specific to methods or engineering approaches. **(ii)** Searched databases more focused on legal, social, and health sciences; not engineering and computer science ones. **(iii)** Topics identified and studied in the SLR barely included health and social care research. **(iv)** Search stops at 2019.
[2]	2023	Yes	Towards Concrete and Connected AI Risk Assessment (C2AIRA): A Systematic Mapping Study	**(i)** Focus on particular risk assessment frameworks for Responsible AI. **(ii)** Address Responsible AI as a broad area, not specific to human rights. **(iii)** Studies selected from gray literature.
[1]	2024	Yes	Applying the ethics of AI: a systematic review of tools for developing and assessing AI-based systems	**(i)** Not focused on human rights, but the broad area of AI Ethics. **(ii)** Uses a broad definition for *Tools*, e.g., including regulations, guidelines, grey literature, scientific papers, etc. **(iii)** Search limited to the years of 2015 to 2022.
[10]	2025	No	Fostering Human Rights in Responsible AI: A Systematic Review for Best Practices in Industry	**(i)** It is not a systematic review of *scientific* literature, but a review of industry-authored resources (e.g., including blog posts, privacy policies, reports, and guidelines). **(ii)** Searches were restricted to four companies (i.e., OpenAI, Google AI, Meta AI Research, Microsoft AI). **(iii)** The proposed best practices are mostly of high-level recommendations (e.g., "engage with NGOs," "collaborate with academia," or "share research findings").
This SMS		Yes	**Main Topic:** Human Rights Engineering Strategies for AI Systems	**(i)** Focus on *human rights*, rather than broader/vaguer concepts of ethical, responsible, or trustworthy AI. **(ii)** Focus on *engineering* strategies (i.e., techniques, methods, processes, frameworks, and tools) for a human rights-based AI design and development.

borders and cultures, ensuring that AI serves shared human values as embodied in the Universal Declaration of Human Rights (UDHR) and other instruments. (b) The international human rights regime is binding on States. (c) Human rights law and jurisprudence provide a framework for balancing rights that conflict with each other. With that in mind, the following section further formulates the protocol for conducting and documenting the SMS.

2.1.2 Specifying the Research Questions

To conduct the mapping study and present its results, we specify some Research Questions (RQs) for this SMS. Answers to the specified RQs represent the SMS results and justify the SMS's scope. In addition, answers to individual RQs help us objectively document the study's results. In the following, we present the RQs along with their main objectives:

A. Systematic Mapping of Studies, Types, Frequencies, and Research Venues

RQ-1 *What is the type of and frequency of published research in the area of Human-Rights-by-Design for AI systems?* Objective(s): To understand and highlight the (i) type of published research in terms of conference proceedings, journal articles, symposium papers, etc.; (ii) frequency of published research that reflects progression and growth of research in terms of the number of publications over the years; and (iii) authors' affiliations and geographical distribution of the research.

RQ-2 *What are the prominent venues of research publications in the area under investigation?* Objective(s): To list and analyze the publication venues (e.g., fields and disciplines), such as specific conference proceedings, journal articles, and special issues, along with book chapters that highlight prominent venues of research emergence that are detailed below.

B. Synthesis of Research, Engineering Solutions, their Evaluations, and Validation Research

RQ-3 *What are the current human rights engineering strategies for designing AI-based digital health solutions, and what are their limitations?* Objective(s): Identification of the proposed solution represents various research themes – reflecting the body of knowledge – that help us to investigate the strengths and limitations of existing research.

RQ-4 *How can the identified approaches be organized in a taxonomy and mapped to various software development processes in the AI lifecycle?* Objective(s): As part of the data synthesis, we seek to provide a systematization of knowledge in the form of a taxonomy that summarizes the identified strategies.

RQ-5 *How are human rights conceptualized in the identified approaches?* Objective(s): Provide a detailed description of how the concept of human rights is articulated in the identified research, delving deeper into the notion through the research team's multidisciplinary legal, technological, and critical humanities lenses.

RQ-6 *What is the state of existing evaluation studies on Human-Rights-by-Design for AI systems?* Objective(s): Provide a clear picture of the existing research that has been properly evaluated and identify the areas of Human-Rights-by-Design for AI systems at the forefront of science.

2.2 Phase II – Conducting the Mapping Study

As per Figure 1, this phase involves searching and qualitative assessment of the relevant literature for review to conduct the SMS, as detailed below.

2.2.1 Search the Relevant Literature

For searching the relevant literature, we select the Scopus database as the digital library indexes more than five thousand publishers, including highly relevant sources such as Elsevier, Springer, MEDLINE, EMBASE, IEEE Xplore, and ACM. To search literature in Scopus, we consider the outlined RQs (from Section 2.1.1) to compose the search string based on the key terms that are presented in Table 2.

Table 2: Key terms used divided by groups.

G1	G2
fundamental right*	artificial intelligence*
human right*	machine learning
basic right*	machine intelligence
natural right*	expert system*
fundamental freedom*	neural network*
	robot*

We limited the search to peer-reviewed scientific publications and book chapters, excluding white papers, technical reports, or unpublished work. However, we did not limit the searches in terms of year ranges. Notice that we also do not add more search terms (e.g., "engineer*" or "developer*") to avoid further restricting the search. This is possible given that the current search string, which focuses on "human rights" and "AI," returns a feasible number of results for the research team to screen.

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TITLE-ABS-KEY(
    ( "fundamental right*" OR "human right*" ) AND
    ( "artificial intelligence" OR "machine learning" )
)
```

Listing 2: Composition of Search String for Literature Search (Scopus: 1968 documents found - Sep 2025).

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TITLE-ABS-KEY(
    ( "fundamental right*" OR "human right*" OR "basic right*"
    OR "natural right*" OR "fundamental freedom*" ) AND ( "artificial
    intelligence" OR "machine learning" OR "machine intelligence"
    OR " expert system*" OR " neural network*" OR "robot*")
)
```

Listing 3: Extended Composition of Search String for Literature Search (Scopus: 2237 documents found - Sep 2025).

2.2.2 Study Inclusion – Screening and Qualitative Assessment

Out of the potentially relevant studies, we need to select the relevant ones to be included for review in the SMS, based on screening and quality assessment as in Table 3.

Step I – Screening of the relevant literature: As in Table 3, Step I consists of a screening using five criteria to include or exclude the relevant literature. The screening is performed by quickly reviewing the titles and abstracts of the identified studies. Two researchers perform this primary selection of papers. When in doubt during the screening process, a third researcher is contacted to resolve the inclusion or exclusion of a paper. The publication is excluded if any criteria from S1 to S5, as in Table 3, resulted in a NO. As a result of the screening, the included publications are moved to Step 2 for the qualitative assessment.

Step II – Quality assessment of screened studies: A quality assessment of the screened studies enables us to perform a criteria-based evaluation according to the guidelines in [4] and the five-point criteria detailed in Table 3. Qualitative assessment criteria helped us to include/exclude studies in a step-wise and objective manner. Specifically, the criteria (Q1 to Q5) enable us to perform quality-based ranking and provide a numerical quantification for individual studies. The maximum score for any study can be 1, and the minimum is 0. Any study with a qualitative assessment score below 0.4 is excluded based on a lack of quality to be included in the review. However, during this screening, we also check for consistency, following the criteria from S1 to S5, as shown in Table 3.

2.2.3 Synthesize Qualitatively Selected Literature

The last task of Phase II is the classification of the studies in systematic maps. Based on that, we also critically appraise the "Evaluation Research," i.e., primary research that evaluates solutions implemented in practice (see Table 4).

Qualitative Assessment and Classification Scheme (Facets): Classification schemes (or facets) contain a set of categories that is representative of the underlying population. In this SMS, we used six facets: (1) research type; (2) human rights addressed; (3) other relevant regulations mentioned; (4) engineering strategy types (e.g., techniques, methods, processes, frameworks, and tools); (5) AI life-cycle process targeted; (6) types of AI/ML used; (7) application contexts (e.g., health, transportation, manufacturing); (8) possible impact assessments of a proposed solution; and (9) intersectionality and gender aspects considered (and other protected characteristics, such as race, age, or religion). Existing facets already proposed in the literature were adopted instead of creating new classification schemes unnecessarily. For research types, the facet proposed in [13] was used (see Table 4). The remaining classification facets will be constructed during the data analysis based on the content extracted from the selected studies and consolidated during the data charting step.

Before starting the mapping process, the classification facets are tested for consistency with a subset of papers (e.g., approx. 10%). In this test, all the papers are classified using these facets, allowing us to verify if they are adequate for the data analysis.

Critical Appraisal and Analysis Focused on Evaluation Research: Building on the initial SMS, an in-depth analysis focuses on the papers categorized as "Evaluation Research" in the research type facet. These are the publications describing technologies that were implemented in practice, and an evaluation of the technique is conducted. This also includes evaluating existing technologies on the market and identifying problems in the industry. Evaluation studies were chosen because they comprise the most advanced stage of the research process. In the evaluation stage, when things are being

Table 3: Inclusion and exclusion criteria.

Step I – Screening (S) of Relevant Literature	**Yes/No**
S1. Is the study in the English language?	
S2. Is the study scientific, peer-reviewed, and published research (no white papers or technical reports, etc.)?	
S3. Is the study not a book or book chapter (i.e., lacks proper peer review)?	
S4. Is the study not a secondary study with no solution proposal (e.g., taxonomy or guidelines)?	
S5. Is the study not solely a discussion or opinion paper (e.g., discusses or examines problems, challenges, or trends)?	
S6. Is the study not a purely rhetorical or theoretical paper, without proposing any engineering solution on the topic?	
S7. Is the study not a purely legal paper, without a clear engineering solution for human rights in the design of AI systems?	
S8. Is the study proposing or validating/evaluating solutions (e.g., techniques, methods, processes, frameworks, and tools) for human rights engineering strategies in AI-based systems?	
S9. Is the proposed solution for the broad integration of human rights into AI design, rather than addressing a right in isolation (e.g., only a privacy-enhancing mechanism, only a fairness metric, only an explainability/interpretability technique)?	
If [Yes] to all nine criteria (S1 to S9) then go to Step II, otherwise exclude the study	
Step II – Quality (Q) Assessment of Screened Studies	**Y/P/N**
Q1. Are the problem definition and proposed solution(s) clearly presented?	
Q2. Is the research environment or context in which the study was carried out properly explained?	
Q3. Are the research methodology and its organization clearly stated?	
Q4. Are the contributions of the study properly validated or evaluated?	
Q5. Are lessons learned, limitations, and future research explicitly mentioned?	

Note: Yes (Y) = 1.0, Partially (P) = 0.5, No (N) = 0.0.

Table 4: Research Type Facet (based on [13]).

Category	**Description**
Validation Research	Techniques investigated are novel and have not yet been implemented in practice. Techniques used are, for example, experiments, i.e., work done in the lab.
Evaluation Research	Techniques are implemented in practice, and an evaluation of the technique is conducted. That means it shows how the technique is implemented in practice (solution implementation) and the consequences of the implementation in terms of benefits and drawbacks (implementation evaluation). This also includes identifying problems in the industry.
Solution Proposal	A solution for a problem is proposed; the solution can be either novel or a significant extension of an existing technique. The potential benefits and the applicability of the solution are shown by a small example or a good line of argumentation.
Philosophical Papers	These papers sketch a new way of looking at existing things by structuring the field in the form of a taxonomy or conceptual framework.
*Opinion Papers	These papers express the personal opinion of somebody on whether a certain technique is good or bad, or how things should be done. They do not rely on related work and research methodologies.
*Experience Papers	Experience papers explain what and how something has been done in practice. It has to be the personal experience of the author.

*: Although the categories of Opinion and Experience papers are part of the research type facet, we do not expect to use them, as we focus on primary research with solution proposals.

implemented in the real world, the bleeding edge of technology reveals itself, allowing professionals to decide on the adoption (or rejection) of new solutions.

To that end, we also seek to identify the methods used in the evaluation research reported in

the literature, in which all papers were read in full and manually annotated. It is worth noting that Human-Rights-by-Design aspects are emphasized in this second-stage analysis, seeking to identify clear evidence of evaluation (or lack thereof) of the approaches claimed to be part of the AI solutions. We also performed a quality assessment for each paper based on the criteria proposed in [14]. A summary of the quality assessment criteria for these papers is presented in Table 5.

Table 5: Quality criteria used in the second-stage analysis (from [14]).

Answers: Yes (1), Partly (0.5), No (0).
1. Is the paper based on research (or is it merely a "lessons learned" report based on expert opinion)?
2. Is there a clear statement of the aims of the research?
3. Is there an adequate description of the context in which the research was carried out?
4. Was the research design appropriate to address the aims of the research?
5. Was the recruitment strategy appropriate to the aims of the research?
6. Was there a control group with which to compare treatments?
7. Was the data collected in a way that addressed the research issue?
8. Was the data analysis sufficiently rigorous?
9. Has the relationship between researcher and participants been considered to an adequate degree?
10. Is there a clear statement of findings?
11. Is the study of value for research or practice?

2.3 Phase III – Documenting the Mapping Study

As per Figure 1, the last phase of the SMS, i.e., **Documenting the Mapping Study**, is detailed in a separate manuscript. The artifacts used and created in this study are publicly available in a replication package. This package can be accessed at [15]. It includes the database search queries, the answer sets of these queries, and the maps of categorized papers and patterns.

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